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Received: 24 April 2026

Accepted: 29 July 2026

Published online: 10 August 2026

Cite this article as: Elesin, L., Shilov, A.L., Kravtsov, M. *et al.* Correlated insulator Moiré bolometer. *Nat Commun* (2026). <https://doi.org/10.1038/s41467-026-76389-4>

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Correlated Insulator Moiré Bolometer

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ABSTRACT

Light incident on an insulator is generally not expected to turn it into a metal without invoking intense ultrafast excitation that leads to transient structural transitions. Here we show that magic-angle twisted bilayer graphene tuned to half filling of the moiré band provides a notable exception to this expectation. We find that weak long-wavelength photons, with energies comparable to the flat-band width, selectively heat the low-heat-capacity electronic subsystem, thereby suppressing the correlated gap. This produces a giant resistance change governed not by a persistent photocarrier population, but by the extreme sensitivity of a many-body correlated gap to weak electronic heating. The resulting photon-driven insulator-to-metal transition produces a broadband low-noise photoresponse with voltage responsivity exceeding millivolts per nW of absorbed power. The mechanism is dual to superconducting hot-electron response: radiation-heated electrons suppress a many-body order, but in reverse the correlated insulator melts into a metal, providing robustness to magnetic fields of several tesla and a sharp insulator-to-metal resistive contrast. Our results establish correlated flat-band systems as a platform for ultra-sensitive detection of faint long-wavelength radiation.

INTRODUCTION

Light incident on a material can alter its electrical resistance – a class of phenomena termed photoresistance¹. In conventional semiconductors, absorbed photons promote electrons across a bandgap, generating free carriers that enhance conductivity. In metals, by contrast, photoresistance is typically negligible: although absorption may heat the electron gas, the large Fermi energy and high carrier density render this perturbation

largely irrelevant for transport. Reducing dimensionality changes this picture qualitatively: resonant excitations in quantum dots, hot-carrier and thermoelectric effects in low-dimensional conductors, and other mechanisms can strongly amplify the impact of optical excitation on transport, giving rise to pronounced photoconductive responses and enabling new device functionalities^{2–7}. Yet a unifying feature of these standard effects is their single-particle character: photons couple to individual electrons, redistributing them in energy or momentum and thereby modifying resistance through changes in carrier density or scattering.

A handful of exceptions illustrate a qualitatively different regime. In ultra-clean low-dimensional conductors near the hydrodynamic regime, light-induced heating can drive a crossover from ballistic to viscous electron flow, producing a photoresponse governed by collective transport coefficients⁹. Furthermore, plasmons – collective self-sustained oscillations of electron density – can resonantly enhance absorption of low-dimensional conductors^{10–13} and thereby can influence the photoresponse. Electromagnetic fields can also directly excite collective charge-order dynamics, as recently demonstrated by atomically resolved THz spectroscopy of charge-density-wave phase excitations in NbSe₂¹⁴. More dramatically, in superconducting thin films and nanowires, absorbed photons readily suppress the zero-resistance state, enabling ultrasensitive photodetection^{15–18}. This example points to a broader principle: when electrical resistance is governed by a many-body order parameter rather than single-particle band structure, photons need only destabilize that order to produce a dramatic response. Correlated two-dimensional quantum materials provide a natural platform to revisit this idea, as their transport properties are defined by emergent electronic order stabilized by interaction energies that can be remarkably small^{19–27}. Yet despite intense fundamental interest in these materials, their optoelectronic behavior almost always^{18,28} has remained confined to single-particle physics^{29–32}. Graphene-based moiré flat-band systems are particularly compelling in this context: their fragile

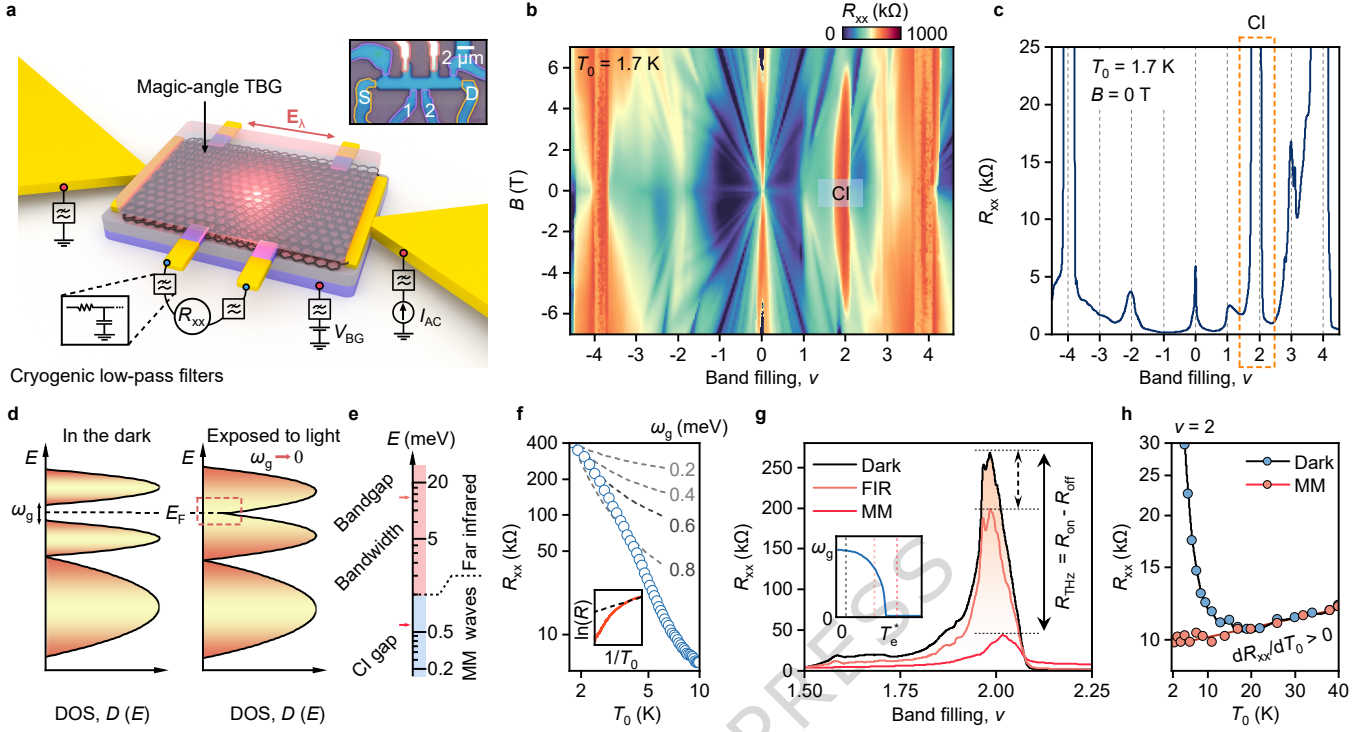


FIG. 1. Correlated Insulator Moiré Bolometer. **a**, Schematic of the magic-angle twisted bilayer graphene (TBG) device. The red spot indicates hot electrons generated in the TBG channel under THz irradiation with electric field $\mathbf{E}_\lambda$, where λ is the radiation wavelength. The longitudinal resistance R_{xx} was measured in a four-probe configuration using an AC excitation current I_{AC} , while the carrier density was tuned by the graphite back-gate voltage V_{BG} . Inset: optical image showing the source (S), drain (D), and voltage probes. **b**, Landau-fan diagram of R_{xx} measured at the cryostat base temperature $T_0 = 1.7$ K, showing a pronounced correlated insulator (CI) state. **c**, R_{xx} versus band filling; the dashed rectangle marks the CI state. **d**, Illustration of the density of states (DOS) in the dark and under irradiation. At half-filling, the Fermi level E_F lies within the correlated gap ω_g . Under sub-THz irradiation, the gap closes, $\omega_g \rightarrow 0$, producing a finite DOS at E_F highlighted by the red dashed box. **e**, Energy scale in the system. **f**, Collapse of the CI state with increasing temperature. Gray dashed lines show Arrhenius fits yielding the corresponding gaps ω_g . Inset: the same data in Arrhenius coordinates, showing behavior consistent with Ref.⁸; the black dashed line is the limiting Arrhenius fit as $T \rightarrow T_0 = 1.7$ K. Data were measured at 500 pA to minimize current-induced overheating. **g**, Comparison of R_{xx} measured in the dark (R_{off}) and under far-infrared (FIR) or millimetre-wave (MM) irradiation (R_{on}). The photoresistance is defined as $R_{THz}(\nu) = R_{on}(\nu) - R_{off}(\nu)$. Inset: qualitative dependence of ω_g on electronic temperature T_e ; the gap vanishes with increasing temperature (T_e^* denotes the critical electron temperature), and dashed lines indicate the radiation-induced increase in T_e . **h**, Temperature dependence of R_{xx} with and without MM-wave illumination at $\nu = 2$. Under illumination, the device shows metallic behavior, $dR_{xx}/dT_0 > 0$. This curve was acquired during a different cooldown; therefore, the absolute resistance values differ from those shown in Fig. 1g.

correlated insulating (CI) states are controlled by energy scales of only a few millielectronvolts – often the smallest scale in the system^{8,33–36}. This raises a fundamental question: can weak, low-energy photons destabilize such an insulator and drive it metallic, producing a giant photoresistive response without invoking structural transitions or lattice dynamics?^{37–39} Until now, this was thought to require the brute force of intense, above-gap femtosecond excitation that injects carriers across eV-scale Mott or charge-transfer gaps^{40–42}, thereby overlooking a regime in which weak perturbations can produce disproportionately large electronic responses.

Here we show that this expectation does not apply to magic-angle twisted bilayer graphene (MATBG) and demonstrate giant photoresponse of MATBG to low-intensity long-wavelength photons when it is tuned to

half filling of the moiré band^{8,33–36}. Far-infrared photons selectively overheat the low-heat-capacity electronic subsystem^{43,44}, driving a collapse of the correlated order and producing a giant photoresistive response. By studying the dependence on band filling, temperature, and radiation power, we show that the effect originates from the destabilization of the many-body gap rather than from single-particle excitation. Beyond its implications for correlated-electron physics, this mechanism enables highly sensitive far-infrared detection with megavolt-per-watt internal responsivity and low noise.

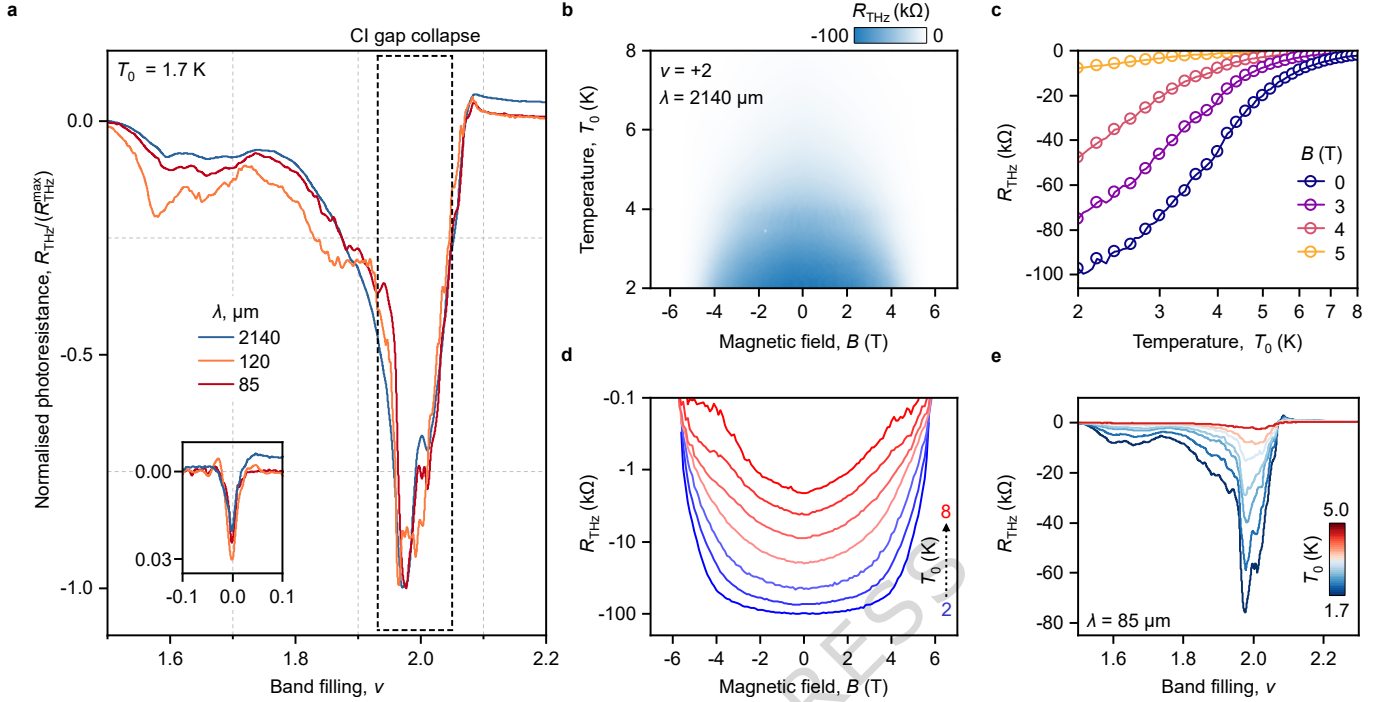


FIG. 2. **Broadband photoresistance of the CI state.** **a**, Normalised photoresistance $R_{\text{THz}}/R_{\text{THz}}^{\text{max}}$ as a function of filling factor ν near the CI state ($\nu = 2$) for different wavelengths. The dashed box highlights the region of the most pronounced negative photoresistance near $\nu = 2$. Inset: normalised photoresistance in the vicinity of the charge-neutrality point (CNP). **b**, Phase diagram of the photoresistance as a function of external magnetic field and temperature. **c**, Temperature-dependent photoresistance traces at selected magnetic fields. **d**, Magnetic-field dependence of photoresistance at the CI state for selected temperatures. **e**, Band-filling dependence of photoresistance under illumination with $\lambda = 85 \mu\text{m}$ radiation at various temperatures.

RESULTS

Correlated Insulator Moiré Bolometer. To explore the bolometric capabilities of the CI state, we fabricated a device based on MATBG, encapsulated in two hexagonal boron nitride (hBN) slabs (see Methods). The device was patterned in the form of a multiterminal field-effect transistor (FET) with a global graphite back gate, enabling precise control of carrier doping of the whole device (Fig. 1a).

Figure 1b presents the longitudinal resistance, R_{xx} , mapped as a function of moiré band filling, $\nu = n/n_0$ (where n is the electrostatically doped carrier concentration and $4n_0$ represents the full filling of superlattice minibands), and perpendicular magnetic field, B , together with its zero-field cross-section (Fig. 1c). The data exhibit the characteristic features of MATBG: a charge-neutrality point at $\nu = 0$, strong single-particle insulating states at $\nu = \pm 4$, and CIs at $\nu = \pm 2$, with the $\nu = +2$ state being particularly pronounced.

Figure 1f displays the temperature dependence of R_{xx} measured at $\nu = 2$. The insulating behavior is remarkably steep: R_{xx} increases by more than an order of magnitude upon cooling the sample from 10 K to 1.7 K. In this temperature window, $R_{xx}(T_0)$ clearly deviates from the simple Arrhenius dependence expected only for a

temperature-independent gap (Fig. 1f)⁸. This behavior is consistent with previous scanning tunneling microscopy studies^{33–35}, which identified that the ground state at $\nu \approx 2$ is a symmetry-broken insulating phase with an intervalley-coherence (IVC) order, characterized by an energy gap of approximately 1 meV below 1 K. This gap is gradually closing upon raising T above 1.5 K and fully vanishes around 10 K^{8,33–35}, consistent with our transport data.

Collapse of the CI and giant photoresistance.

Figure 1g demonstrates the extreme sensitivity of the sample resistance at $\nu = 2$ to incident millimetre-wave (MM) and far-infrared (FIR) photons. Photon absorption heats the electronic subsystem, destabilizing the fragile CI state and reducing the sample resistance. This sensitivity arises from the following mechanism.

External illumination increases the effective carrier temperature in the flat band (see the results of the independent Johnson noise thermometry measurements in Supplementary Figure 12). This, in turn, leads to the destabilization of the CI, restoring a finite density of states near the Fermi level (see Figure 1d) and producing a dramatic drop in resistivity. The effect is conceptually symmetric to that of superconducting detectors. In both cases, electron heating perturbs a correlated electronic phase; however, while superconducting bolometers rely

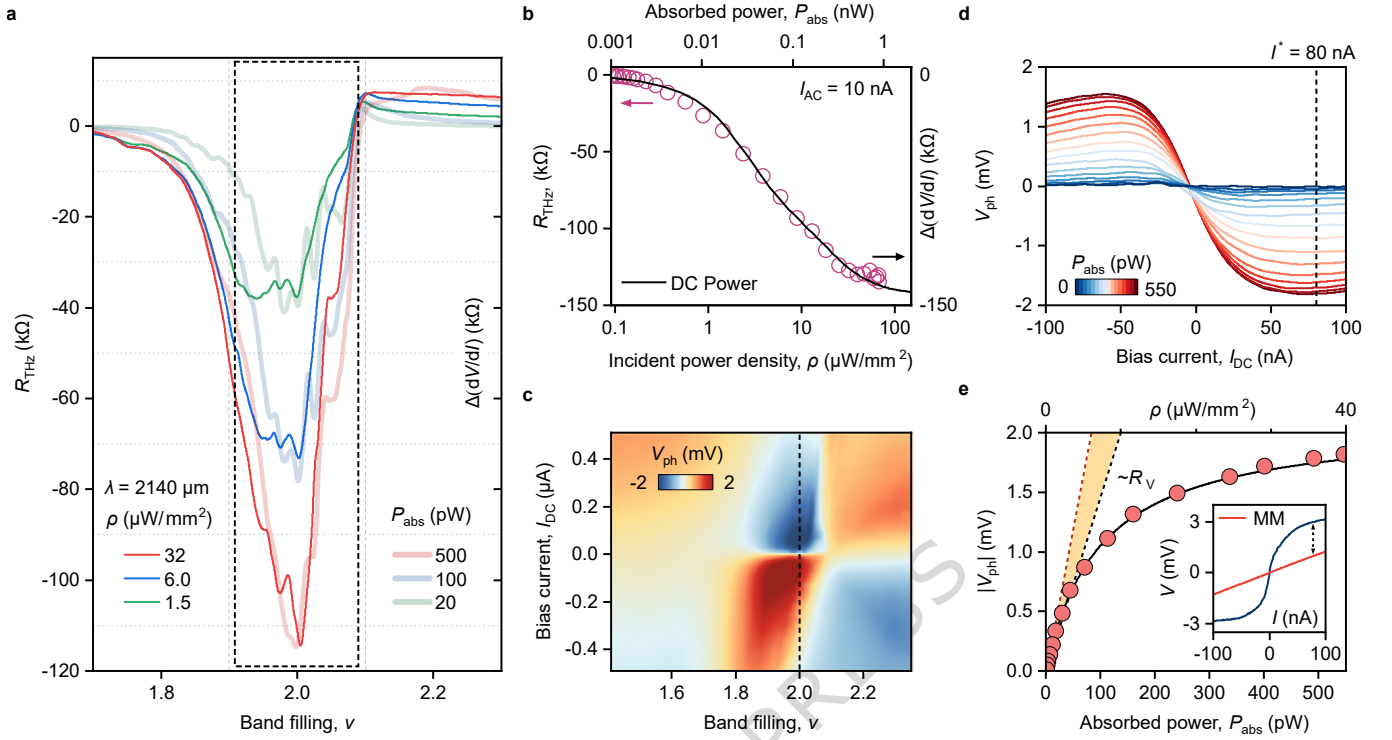


FIG. 3. Responsivity of the CI moiré bolometer. Photoresponse data shown in this figure were acquired at $\lambda = 2140 \mu\text{m}$. **a**, Comparison of the THz-induced photoresistance $R_{\text{THz}}(\nu)$ near the CI with the change in differential resistance $dV/dI(P_{\text{abs}}) - dV/dI(0)$ produced by an equivalent absorbed power P_{abs} generated via DC heating from a DC bias current I_{DC} . The dashed box highlights the selected region near $\nu = 2$. **b**, Comparison between the photoresistance at $\nu = 2$ as a function of incident THz power density and the change in $dV/dI(P_{\text{abs}}) - dV/dI(0)$ produced by DC heating with power P_{abs} . **c**, Photovoltage V_{ph} mapped as a function of ν and DC bias I_{DC} . **d**, V_{ph} vs bias current I_{DC} under THz radiation illumination of different power (the incident power is recalculated to absorbed power using the calibration curve). The dashed line represents the optimal operation point I^* of the bolometer. **e**, $|V_{\text{ph}}|$ as a function of P_{abs} under THz illumination. The data were acquired at a fixed DC bias current of $I^* = 80 \text{ nA}$. Dashed lines represent the responsivity of the device in the linear regime R_V , and the solid line is a saturation fit $|V_{\text{ph}}| = V_0 \frac{P_{\text{abs}}}{P_0 + P_{\text{abs}}}$ (for $V_0 = 2.1 \text{ mV}$ and $P_0 = 100 \text{ pW}$). Inset: V - I characteristics under MM-wave irradiation and in the dark: the difference in voltage at fixed current corresponds to the photovoltage.

on the destruction of a zero-resistance state, our MATBG bolometer exploits photon-driven CI to metal transition. The R_{xx} dependence on T_0 corroborates this interpretation by comparing the data taken in the dark and under MM-wave excitation (Fig. 1h). While the former signals insulating temperature dependence, the latter reveals a clear $dR_{\text{xx}}/dT_0 > 0$ trend – an indication of a metallic state (See Supplementary Figure 17 for the full phase diagram). Since the CI gap constitutes the smallest energy scale in the system (Fig. 1e), the bolometric response extends over a broad spectral range, encompassing the technologically important MM-wave and FIR bands.

To facilitate further analysis, we define the photoresistance as $R_{\text{THz}}(\nu) = R_{\text{on}}(\nu) - R_{\text{off}}(\nu)$, where R_{off} and R_{on} denote resistance values recorded in the dark and under illumination, respectively (see Methods). Figure 2 summarizes the photoresponse characteristics of the MATBG bolometer. The normalised photoresistance traces $R_{\text{THz}}/R_{\text{THz}}^{\text{max}}$ in Fig. 2a (where $R_{\text{THz}}^{\text{max}}$ is the maximum photoresistance at a given wavelength and power) reveal a pronounced response of the CI state at $\nu = 2$

to incident radiation over a wide spectral range, from $\lambda = 85 \mu\text{m}$ to $\lambda = 2140 \mu\text{m}$. In contrast, outside the CI the system exhibits a finite and positive R_{THz} , consistent with metallic behavior (Supplementary Figure 1e shows quantum oscillations in the resistance as a function of magnetic field for $\nu > 2.1$, indicating the presence of a well-defined Fermi surface).

Figure 2a reveals the absence of distinct spectral features across this photon energy range. This broadband character indicates that the photoresistance response is governed by electronic heating, transduced through the strong temperature dependence of R_{xx} , rather than by a resonant process tied to a specific photon energy. We emphasize, however, that the flatness of the spectral response constrains the transduction mechanism but does not by itself identify the microscopic absorption pathway: interband absorption followed by ultrafast carrier thermalization⁴⁵ deposits energy into the electronic subsystem and would produce an equivalent bolometric signal⁴⁶. In either case, the absorbed power raises T_e and suppresses the correlated order. Furthermore, to

test the coupling mechanism, our device was endowed with a broadband bow-tie antenna designed for linearly-polarized light. We found that the measured R_{THz} is independent of the incident radiation polarization (see Supplementary Figures 5 and 6), suggesting that the response arises from direct absorption⁴⁷ (see Supplementary Notes 3, 4 and 12).

As temperature increases (Fig. 2b–e), the photoresistance amplitude decreases gradually but remains observable up to $T_0 \approx 6\text{--}7\text{ K}$, demonstrating that the bolometric effect persists well above the base temperature of 1.7 K. Furthermore, measurements in perpendicular magnetic fields reveal that the photoresponse remains robust up to several tesla, with substantial reduction occurring only above 4 T.

We have also measured R_{THz} as a function of radiation power density, ρ . Upon raising the latter, the maximum R_{THz} at the CI state increases systematically, consistent with the elevated electronic temperature change ΔT_e (shown in Fig. 3a). In principle, one can use DC heating that allows us to generate the same ΔT_e in a controlled manner: by sweeping I_{DC} , we gradually raise T_e and monitor the resulting suppression of the differential resistance $\frac{dV}{dT}(P_{\text{abs}})$, where P_{abs} is the absorbed power. Figure 3a displays the filling-factor dependence of the R_{THz} alongside the quantity $\frac{dV}{dT}(P_{\text{abs}}) - \frac{dV}{dT}(0)$ with each trace taken at fixed DC heating power P_{abs} . This provides a direct measure of the absorbed power in the detector, a standard approach in bolometric techniques^{9,48}. The close correspondence between these two datasets demonstrates that both mechanisms – THz absorption and DC heating – modify the resistance through the same underlying electronic temperature increase.

Performance. Conventional bolometers are typically operated in the current or voltage bias regimes, so that instead of relating the incident power to the photoresistance, one measures $V_{\text{ph}} = I_{\text{DC}} \cdot R_{\text{THz}}$. Thus, to evaluate the bolometric performance under operating conditions, we measure V_{ph} as a function of I_{DC} in proximity to the CI state (Fig. 3c) at the laser modulation frequency (see Methods). By such measurements, one extracts the difference between the I – V curves measured in the dark and under THz excitation (Inset of Fig. 3e). The strongest photovoltage response, as expected, emerges at $\nu = 2$, where the CI exhibits the greatest temperature sensitivity, and the optimal operating point occurs at a bias current of $I^* = 80\text{ nA}$, as evident from the trace of $V_{\text{ph}}(I_{\text{DC}})$ at $\nu = 2$ (Fig. 3d).

Using the power calibration curve from Fig. 3b, we next plot $|V_{\text{ph}}|$ as a function of absorbed power P_{abs} (Fig. 3e). In the linear regime, the voltage responsivity reaches $R_V^{\text{abs}} = (1.7 \pm 0.4) \cdot 10^7\text{ V/W}$, highlighting the extreme intrinsic sensitivity of the CI to the T_e increase (see also Supplementary Figure 3). For practical comparison, however, it is also instructive to estimate the external responsivity, defined with respect to the incident power. Since it is hard to estimate the absorption area, we use the responsivity per power density as the main

practical figure of merit for the current device, which reaches a value of $2 \cdot 10^{-4}\text{ V} \cdot \text{mm}^2/\mu\text{W}$, and potentially could be substantially improved through better absorption engineering (see the noise-equivalent power (NEP) estimate and full detector characterization in Supplementary Table 1). We further note that although electron heating should occur in any TBG device, such a remarkably high internal responsivity is expected to be observed only in high-quality stacks. This is because the fragile CI state must exhibit a pronounced insulating behavior, which varies significantly from device to device (see Supplementary Notes 6 and 7 for the response at filling factor $\nu = -2$ and measurements on additional MATBG devices, respectively).

Finally, we consider the response time of the CI bolometers. In MATBG, carrier thermalization was experimentally shown to be governed by ultrafast electron–phonon Umklapp cooling, which establishes a picosecond intrinsic timescale for the photoresponse⁴⁵. In the Supplementary Information, we demonstrate non-linear dependence of electron overheating on the absorbed power, indicating electron–phonon cooling as the dominant heat escape mechanism. In our devices, however, the experimentally accessible speed is likely limited by the electrical RC time constant, τ_{RC} , due to the high device resistance. Because the structure is fabricated on an insulating substrate, the dominant capacitance originates from the graphene–graphite back-gate geometry. Using a typical operating resistance $R \sim 200\text{ k}\Omega$ at the point I^* and estimating the capacitance as $C \approx 0.25\text{ pF}$ – assuming an upper-bound device area of $250\text{ }\mu\text{m}^2$ including the gated graphite leads – we estimate $\tau_{\text{RC}} \sim 50\text{ ns}$ that in principle can be reduced by at least an order of magnitude in smaller devices. This value therefore sets the practical upper limit on the response speed in the present device geometry.

DISCUSSION

Our experiments can be further generalized for time-domain interrogation of the CI state itself. Radiation-driven excitation confined within the flat band, combined with time-resolved monitoring of the CI photoresistance, would allow direct access to the intrinsic timescale of correlated-order suppression and recovery, thereby probing the collective dynamics within the ground state. Unlike previous optical studies relying on high-energy interband excitation, such measurements would isolate the response of the many-body order parameter to low-energy perturbations and provide a new route to studying collective modes in moiré quantum materials. In addition to fundamental importance, the demonstrated internal responsivity exceeds that of many commercial bolometers in the same spectral window⁴⁹. Depending on the wavelength, integration with cavities or metamaterials, plasmonic resonators, or photonic-crystal architectures, routinely employed in the field, could raise absorption

toward unity and plausibly elevate responsivities beyond 10^7 V/W without compromising the nanosecond-scale response. Moreover, unlike superconducting detectors, such devices remain operational in magnetic fields up to several tesla, enabling compatibility with magneto-optical instrumentation. Looking forward, alternative correlated phases at different fillings, multilayer twisted stacks, their wafer-scale growth⁵⁰, and integrated cavity geometries may support both single-pixel sensors and large-format arrays.

METHODS

Sample fabrication. Our samples were fabricated from two graphene flakes cut using an AFM tip and subsequently encapsulated between hBN slabs using a standard dry-transfer technique⁵¹. The relative twist angle between the two graphene layers was set during assembly with the aid of an optical microscope equipped with micromanipulators and a high-precision rotation stage. An additional graphite layer was attached to the TBG to serve as a graphite contact for the TBG. Narrow graphite was attached to the bottom surface of the heterostructure to serve as a local back gate. The completed stack was released onto an undoped, insulating Si/SiO₂ substrate to suppress reflections of incident THz radiation. Finally, standard electron-beam lithography, reactive ion etching, and thin-film metal deposition were used to define the contact leads.

Transport and photoresponse measurements. The samples were measured in a Quantum Design Opti-Cool magneto-optical cryostat with a base temperature $T_0 = 1.7$ K. Terahertz radiation at 0.14 THz was generated by a Terasense source, while higher frequencies (2.5 and 3.5 THz) were produced using a quantum cascade laser (Lytid QCL). The radiation was guided to the device via an optical path consisting of two lenses, two mirrors, and a beam splitter, with the final lens mounted on a motorized XYZ stage to optimize focusing using the photoresponse signal as feedback (see Supplementary Figure 2).

Low-noise transport measurements were performed using standard lock-in amplifiers. All electrical contacts were equipped with cryogenic low-pass RC filters to suppress noise originating from the measurement electronics. An AC excitation current, $I(t) = I_{AC} \sin(2\pi f_1 t)$, was supplied by a low-noise voltage-controlled current source, where $I_{AC} = 10$ nA is the current amplitude and $f_1 = 4$ Hz is the excitation frequency. The DC bias (voltage and/or current) and the gate voltage were applied using a low-noise source-meter.

Carrier concentration of our device was controlled by

the electrostatic back gate, where n is given as:

$$n = \frac{C_{BG}}{e} (V_{BG} - V_{BG}^0) \quad (1)$$

where C_{BG} is the bottom-gate capacitance per unit area; V_{BG}^0 denotes the corresponding charge-neutrality voltage, where e is the elementary charge.

The photoresistance was defined as

$$R_{THz}(\nu) = R_{on}(\nu) - R_{off}(\nu). \quad (2)$$

Rather than measuring $R_{on}(\nu)$ and $R_{off}(\nu)$ separately, we employed a dual-modulation technique^{9,52} in which the THz radiation was square-wave modulated at $f_{mod} = 9$ Hz while simultaneously modulating the current at f_1 . The measured voltage can be written as

$$V(t) = I(t)R_{xx}(t) + V_{ph}(t), \quad (3)$$

where $V_{ph}(t)$ is the photovoltage oscillating at f_{mod} . Since the electronic temperature relaxation time is much shorter than $1/f_{mod} = 111$ ms, both R_{xx} and V_{ph} follow the modulation quasi-instantaneously.

The signal was analyzed using a lock-in amplifier operating in dual mode at the difference frequency $f_{dm} = f_{mod} - f_1$. Fourier decomposition of the square-wave-modulated resistance yields

$$V_{dm} = \frac{I_0 \Delta R_{xx}}{\pi}, \quad (4)$$

allowing direct extraction of the photoresistance,

$$\Delta R_{xx} = \pi \frac{V_{dm}}{I_0}. \quad (5)$$

Photovoltage measurements were performed under DC bias using lock-in detection at f_{mod} . The peak photovoltage was obtained from the RMS lock-in output $V_{ph}^{lock-in}$ via

$$V_{ph} = \frac{\pi}{\sqrt{2}} V_{ph}^{lock-in}, \quad (6)$$

accounting for square-wave modulation and RMS-to-peak conversion.

DATA AVAILABILITY

The data shown in Figs. 1–3 are provided in the Source Data files. All other data that support the findings of this study are available from the corresponding author upon request.

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ACKNOWLEDGEMENTS

We acknowledge Ryo Mizuta Graphics as the source of the 3D optical-component graphics used in Supplementary Figure 2a.

FUNDING STATEMENT

This work is supported by the National Research Foundation, Singapore under its NRF Fellowship (NRF Award NRF-NRFF17-2025-0007 given to D.A.B.). K.S.N. is grateful to the Ministry of Education, Singapore (Research Centre of Excellence award to the Institute for Functional Intelligent Materials, I-FIM, project number EDUNC-33-18-279-V12) and to the Royal Society (UK, grant number RSRP R 190000) for support. K.W. and T.T. acknowledge support from the JSPS KAKENHI (Grant Numbers 21H05233 and 23H02052), the CREST (JPMJCR24A5), JST and World Premier International Research Center Initiative (WPI), MEXT, Japan.

AUTHOR CONTRIBUTIONS

D.A.B. conceived and supervised the project. L.E., A.L.S., M.K., X.Z., and S.J. performed photoresponse measurements. L.E., A.L.S. and S.J. contributed to the fabrication of devices. M.L., A.K. performed antenna absorption simulations. R.I., K.S., and I.G. performed noise thermometry measurements. L.E., A.L.S., M.K. analyzed the data with input from D.A.B.. L.E. and D.A.B. wrote the paper with input from all authors. G.N.G. and K.S.N. provided experimental support. A.P., and I.I. provided theory support. T.T. and K.W. provided high-quality hBN crystals.

COMPETING INTERESTS

The authors declare no competing interests.